

Codebase Genius: AI-Powered Documentation Generator

Author: Evans Langat

Date: October 2025

1. Introduction

Codebase Genius is an AI-driven system that automatically generates comprehensive documentation for GitHub repositories.

2. Objective

- Automate documentation generation.
- Improve maintainability and knowledge transfer.
- Serve as foundation for Jaseci-based intelligent systems.

3. Workflow

1. Accept repository input and validate it.
2. Map repository structure and analyze files.
3. Prioritize main modules (main.py, app.py).
4. Generate Markdown documentation.

4. Implementation

Modules:

- code_genius.py – core logic
- __main__.py – supervisor interface
- output/DOCUMENTATION.md – generated result

5. Results

Generated documentation successfully for multiple Python files.

Output saved in ~/codebase_genius/output/.

6. Challenges and Future Work

Challenges

- Parsing large repos.
- Handling syntax errors.

Future Improvements

- Add Jac graph walkers.
- Generate relationship diagrams.
- Build web/API interface.

7. Conclusion

Codebase Genius proves automated documentation is achievable and sets the stage for Jasoci-powered extensions.